ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** an electric cell	A cell has chemicals that produce electric current.
2	**(b)** conductors	E.g. copper.
3	**(b)** hot	Heating effect of current.
4	**(c)** battery	A battery is a combination of cells.
5	**(c)** insulators	E.g. rubber.
6	**(d)** electric iron	Heating element.
7	**(c)** electric circuit	A complete path is needed.
8	**(b)** copper	Metals conduct.
9	**(d)** nichrome wire	High resistance wire that gets very hot.
10	**(c)** a closed circuit	The bulb glows.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A complete path for the flow of current (1); It has a source, a switch and a device (1).	1+1
12	Connect positive of one to negative of the next (1); Cells joined in series (1).	1+1
13	Cell: long and short parallel lines (1); Bulb: circle with a cross (1).	1+1
14	Closed: current flows (1); Open: gap, no current (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The circuit is open (1); No current flows (1); The filament does not heat (1).	1+1+1
16	Wire melts on excess current (1); Circuit breaks (1); Appliance saved (1).	1+1+1
17	Voltages add (1); Bulb glows brighter (1); Battery formed (1).	1+1+1
	OR: Rubber is an insulator (1); Circuit is not complete (1); No current (1).	1+1+1
18	It gets very hot when current passes (1); It has a high melting point (1); It does not burn or rust easily (1).	1+1+1
	OR: Insulation worn (1); Short circuit (1); Shock or fire (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Make a circuit with a gap (1); Place objects in the gap (1); Copper wire, iron nail, coin: bulb glows (1); Rubber, plastic: no glow (1); Record in a table (1).	1+1+1+1+1
	OR: Battery is two or more cells (1); Join + of one to - of next (1); 1.5 x 3 = 4.5 V (1); More voltage (1); Brighter bulb (1).	1+1+1+1+1
20	Cell: gives energy (1); Wires: path (1); Switch: control (1); Bulb: glows (1); Closed circuit needed (1).	1+1+1+1+1
	OR: Conductors allow current (1); Insulators do not (1); Copper, aluminium (1); Rubber, plastic (1); Uses in wires and covers (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
- Writing that the fuse is connected in parallel with the appliance; it is connected in **series**.
- Mixing up the heating effect (heat from current) with the cause of the glow of a bulb; the filament glows because it becomes very hot.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** plastic	Plastic does not conduct.
2	**(b)** fuse	It protects appliances.
3	**(d)** an open circuit	No current flows.
4	**(c)** graphite	Graphite conducts.
5	**(a)** low melting point	It melts easily.
6	**(c)** switch	A switch makes or breaks the path.
7	**(b)** is an insulator	It protects from shocks.
8	**(b)** MCB	Miniature circuit breaker.
9	**(c)** filament	It is made of tungsten.
10	**(c)** a long line and a short thick line	Long thin = positive, short thick = negative.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A device to open or close a circuit (1); To control the appliance (1).	1+1
12	Current heats a conductor (1); Used in heaters and irons (1).	1+1
13	Conductors: copper, aluminium (1); Insulators: rubber, plastic (1).	1+1
14	It has a very high melting point (1); It glows without melting (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell connected to bulb with wires (1); Switch in the path (1); Close the switch (1).	1+1+1
16	Electric iron (1); Heater (1); Geyser (1).	1+1+1
17	Computer heating (1); Mobile phone (1); Wires overheating (1).	1+1+1
	OR: Do not touch with wet hands (1); Use proper plugs (1); Do not overload sockets (1).	1+1+1
18	Fuse wire melts, must be replaced (1); MCB trips and can be reset (1); Both protect (1).	1+1+1
	OR: Battery gives more voltage (1); More current flows (1); Filament glows brighter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Current heats a wire (1); Iron, heater (1); Special wires used for heating (1); Wires may overheat (1); Cause of fire (1).	1+1+1+1+1
	OR: Glass bulb (1); Tungsten filament (1); Inert gas (1); Current heats filament (1); It glows (1).	1+1+1+1+1
20	Insulated wires (1); Fuse and MCB (1); Switch off before repairs (1); Dry hands (1); Avoid overloading (1).	1+1+1+1+1
	OR: Short circuit: wires touch (1); Overload: too many appliances (1); Large current flows (1); Fuse or MCB (1); Careful wiring (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
- Writing that the fuse is connected in parallel with the appliance; it is connected in **series**.
- Mixing up the heating effect (heat from current) with the cause of the glow of a bulb; the filament glows because it becomes very hot.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** fire	Too much current heats wires.
2	**(b)** current passes through its filament	Current heats the filament.
3	**(d)** a circle with a cross inside	It shows a lamp.
4	**(c)** two bare wires of the circuit touch each other	A path of low resistance forms.
5	**(a)** positive terminal	Marked +.
6	**(a)** gap with a slanted line	It breaks the circuit.
7	**(b)** only one direction	It has polarity.
8	**(b)** negative terminal	Marked -.
9	**(b)** circuit diagram	It shows connections.
10	**(c)** uses much less energy	It saves power.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water conducts electricity (1); You may get a shock (1).	1+1
12	A safety device with a low-melting wire (1); It breaks the circuit when current is too high (1).	1+1
13	Copper is a good conductor (1); It is flexible (1).	1+1
14	Miniature circuit breaker (1); Switches off the circuit on overload (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	They are insulators (1); Prevent shocks (1); Safe use (1).	1+1+1
16	It will not melt on a small overload (1); Wires may overheat (1); Fire risk (1).	1+1+1
17	Fuse (1); MCB (1); Insulating gloves (1).	1+1+1
	OR: Connect cell, bulb and two free ends (1); Touch object across the ends (1); Bulb glows for a conductor and not for an insulator (1).	1+1+1
18	LED has no filament (1); Uses less power (1); Works in one direction (1).	1+1+1
	OR: Contains salty water (1); Current may flow (1); So avoid live wires (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fuse wire in a holder (1); Melts on high current (1); MCB: switch trips (1); Reset easily (1); Protect the circuit (1).	1+1+1+1+1
	OR: The heating element is made of nichrome which gets very hot (1); The copper cord is a good conductor and stays cool (1); The same current flows through both (1); Different materials heat differently (1); So the cord stays cool (1).	1+1+1+1+1
20	Symbols simplify (1); Cell, battery (1); Bulb, switch (1); Connected in series (1); Closed switch lights the bulb (1).	1+1+1+1+1
	OR: Use less power (1); Last longer (1); Little heat (1); Less electricity bill (1); Reduce fossil fuel use (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
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